



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

NEUTRALIZATION AGENT INJECTION PUMPS (P-2306A/B)

Docu. No. : 14780-8120-PS-P2306
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

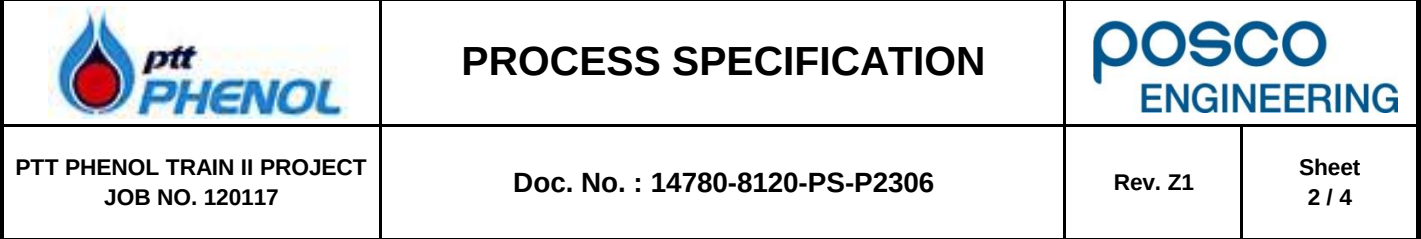
NEUTRALIZATION AGENT INJECTION PUMPS

(P-2306A/B)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-P2306
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Record of Revision	
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PROPORTIONING PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 503

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	14-Aug-13	MW	MEH				

REV

APPLICABLE TO: ☐ PROPOSAL ☐ PURCHASE ☐ AS BUILT

FOR _____ UNIT _____

SITE _____ MODEL _____ NO. OF PUMPS REQUIRED 2

SERVICE Neutralization Agent Injection Pump SIZE AND TYPE _____

MANUFACTURER _____ SERIAL NO. _____

NOTE: ☐ INDICATES INFORMATION TO BE COMPLETED BY PURCHASER ☐ BY MANUFACTURER

GENERAL

NO. MOTOR DRIVEN 2 OTHER DRIVER TYPE _____

PUMP ITEM NO'S P-2306A/B PUMP ITEM NO'S _____

MOTOR ITEM NO'S _____ DRIVER ITEM NO'S _____ GEAR ITEM NO'S _____

MOTOR PROVIDED BY _____ DRIVER PROVIDED BY _____ GEAR PROVIDED BY _____

MOTOR MOUNTED BY _____ DRIVER MOUNTED BY _____ GEAR MOUNTED BY _____

MOTOR DATA SHEET NO. _____ DRIVER DATA SHEET NO. _____ GEAR DATA SHEET NO. _____

☐ OPERATING CONDITIONS

☐ CAPACITY @ PT 1/h

MAXIMUM 6.65 MINIMUM _____ RATED _____

☐ DISCHARGE PRESSURE kg/cm²[g]

MAXIMUM 4.788 MINIMUM _____

☐ SUCTION PRESSURE kg/cm²[g]

MAXIMUM 10.21 MINIMUM _____ .004783

☐ DIFFERENTIAL PRESSURE kg/cm²

MAXIMUM 4.78 MINIMUM _____

☐ NPSH AVAILABLE m 3.0 +

WITHOUT ACCELERATION HEAD _____ ACTUAL _____

☐ TURNDOWN RATIO _____

☐ LIQUID

☐ TYPE OR NAME OF LIQUID Diamine (Dytek A)

☐ PUMPING TEMPERATURE °C

NORMAL 30 MAX _____ MIN _____

☐ SPECIFIC GRAVITY 0.856 MAXIMUM _____ MINIMUM _____

☐ SPECIFIC HEAT 0.46 Cp kcal/(kg.°C)

☐ VISCOSITY cP 2.280 MINIMUM _____ MAXIMUM _____

☐ VAPOR PRESSURE kg/cm²[a] .01000

☐ CORROSIVE/EROSIVE AGENTS _____

☐ CHLORIDE CONCENTRATION (PPM) _____

☐ H₂S CONCENTRATION (PPM) _____

LIQUID ☒ TOXIC ☐ FLAMMABLE ☐ OTHER _____

☐ PERFORMANCE

☐ NUMBER OF FEEDS _____ RATED CAPACITY 1/h

☐ NPSH REQUIRED m

☐ kW RATED _____ AT RELIEF SETTING _____

☐ PLUNGER SPEED (strokes/min) _____ * DESIGN MAX _____

☐ DIAMETER mm LENGTH OF STROKE mm

PUMP HEAD: _____

MAXIMUM PRESSURE kg/cm²[g]

☐ HYDRO TEST PRESSURE kg/cm²[g]

☐ MAX DISCH PRESS. W/ JOB DRIVER kg/cm²[g]

☐ MAX kW BASIS GEAR STRENGTH _____

☐ SITE AND UTILITY DATA

LOCATION ☐ INDOOR ☒ OUTDOOR

☐ HEATED ☒ UNHEATED ☐ UNDER ROOF

☐ ELECTRICAL AREA CLASS 1 GROUP IIA DIV ZONE 2

☐ WINTERIZATION REQD ☐ TROPICALIZATION REQD

SEE REMARKS BELOW ***

SITE DATA

☐ RANGE OF AMBIENT TEMPS:MIN/MAX _____ / _____

UNUSUAL CONDITIONS

☐ DUST ☐ FUMES ☐ SALT ATMOSPHERE

☐ OTHER _____

☐ UTILITY CONDITIONS

☐ CONSTRUCTION

CONNECTIONS

SUCTION

DISCHARGE

FLUSH

LIQUID END ☐ JACKET REQ'D

☐ TYPE ☒ DIAPHRAGM ☐ PLUNGER

☐ DIAPHRAGM DIA mm NO REQ. _____

VALVES/FEED SUCTION DISCHARGE

TYPE _____

NUMBER _____

ELECTRICITY DRIVERS HEATING CONTROL SHUTDOWN

VOLTAGE _____

HERTZ _____

PHASE _____

COOLING WATER INLET RETURN DESIGN MAX

TEMP _____

PRESS. _____

SOURCE _____

INSTRUMENT AIR MAX MIN

PRESSURE _____

APPLICABLE SPECIFICATIONS:

☒ API 675 POSITIVE DISPLACEMENT PUMPS - CONTROLLED VOLUME

☐ GOVERNING SPECIFICATION (IF DIFFERENT)

REMARKS * Per UOP Standard Specification 5-13

No copper or copper bearing alloys shall come in contact with the pumped fluid.

*** See Basic Engineering Design Questionnaire (BEDQ), Engineering Design Information (EDI) Project 963763 A.4, Sect. 3 & 4.

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SHEET 4 / 4

REV	DATE	BY	APVD	REV	DATE	BY	APVD
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REV

MATERIALS

LIQUID END 316 Stainless Steel
 CONTOUR PLATE _____
 HYDRAULIC DIAPHRAGM TFE
 PROCESS DIAPHRAGM TFE
 PLUNGER _____
 LANTERN RING _____
 PACKING GLAND _____
 PACKING _____
 VALVE Ball Type; Stainless Steel, Ceramic or Recommend
 VALVE SEAT Stainless
 VALVE GUIDE Stainless
 VALVE BODY 316 Stainless Steel
 VALVE GASKET _____
 FRAME _____
 SPECIAL MATERIAL TESTS (2.13.1.3) _____
 MINIMUM DESIGN METAL TEMPERATURE °C 15
 LOW AMBIENT TEMPERATURE MATERIALS TESTS (2.13.5) _____

QA INSPECTION AND TEST

- ☐ COMPLIANCE WITH INSPECTORS CHECK LIST
☐ CERTIFICATION OF MATERIALS
☐ FINAL ASSEMBLY CLEARANCES
☐ SURFACE AND SUBSURFACE EXAMINATIONS
 ☐ RADIOGRAPHY _____
 ☐ ULTRASONIC _____
 ☐ MAGNETIC PARTICLE _____
 ☐ LIQUID PENETRANT _____
☐ CLEANLINESS PRIOR TO FINAL ASSEMBLY
☐ HARDNESS OF PARTS, WELDS & HEAT AFFECTED ZONES
☐ FURNISH PROCEDURES FOR OPTIONAL TESTS

TESTS	REQ'D	WIT	OBS
HYDROSTATIC	☒	☐	☐
STEADY STATE ACCURACY	☒	☐	☐
REPEATABILITY	☒	☐	☐
LINEARITY	☒	☐	☐
_____	☐	☐	☐
_____	☐	☐	☐

LUBRICATION FLUID

- ☐ CRANKCASE ☐ INTERMEDIATE
☐ HYDRAULIC FLUID

ACCESSORIES

- ☐ SPEED REDUCER MANUFACTURER _____
☐ INTEGRAL ☐ SEPARATE
 MODEL _____
 RATIO _____
☐ BASEPLATE UNDER _____
☐ COUPLING MANUFACTURER _____
☐ TYPE _____

CONTROLS

TYPE:
☐ MANUAL ☒ REMOTE ☐ PNEUMATIC
☒ AUTOMATIC ☐ LOCAL ☒ ELECTRONIC
 STROKE CONTROL:
 PNEUMATIC kg/cm²[g]
 MINIMUM _____ MAXIMUM _____
 ELECTRONIC (mA):
 MINIMUM _____ MAXIMUM 20

OTHER PURCHASE REQUIREMENTS

- NAMEPLATE UNITS ☐ U.S. CUSTOMARY ☐ SI
☐ VENDOR FURNISHED PROCESS PIPING _____
☐ VENDOR REVIEW PIPING DRAWINGS
☐ VENDOR FURNISHED PULSATION SUPPRESSION DEVICES
☐ VENDOR FURNISHED RELIEF VALVE
☐ INTERNAL ☒ EXTERNAL
☒ RELIEF VALVE SETTING kg/cm²[g] 6.546
☐ VENDOR FURNISHED BACK-PRESSURE VALVE
☐ DOUBLE CHECK VALVES REQUIRED
☐ OIL-FILLED PRESSURE GAUGES REQUIRED
☐ VENDOR FURNISHED CONTROL PANEL
☐ BASEPLATE PREPARED FOR EPOXY GROUT
☐ PROVIDE TECHNICAL DATA MANUAL
☐ Relief valve provided by others
☐
☐

PREPARATION FOR SHIPMENT

- ☐ DOMESTIC ☐ EXPORT ☐ EXPORT BOXING
☐ OUTDOOR STORAGE MORE THAN 6 MONTHS

WEIGHTS (LBS)

- ☐ PUMP _____ ☐ BASE _____ ☐ GEAR _____ ☐ DRIVER _____

DRIVERS

- ☐ MOTOR:
☐ MANUFACTURER _____
☐ TYPE _____
☐ FRAME NO. _____
☐ CONSTANT SPEED _____
☐ VARIABLE SPEED _____
☐ kW _____ RPM _____
☐ VOLTS _____ PHASE _____
☐ HERTZ _____ SERVICE FACTOR _____
☐ ENCLOSURE _____
☐ OTHER (SEE SEPARATE DATA SHEETS) _____
☐ GAS DRIVEN _____
☐ STEAM TURBINE _____
☐ OTHER _____

See UOP Standard Specification 7-12

REMARKS

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

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